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Looking back on the UniSquare project as a whole, I can honestly say this has been one of the most challenging but rewarding software development experiences I have had so far. Throughout the project, and especially during Sprint 4, I mainly focused on frontend development, UI/UX improvements, documentation, and helping maintain team organization.

A large part of my contribution involved redesigning and improving the homepage/welcome page of UniSquare. I really enjoyed working on this because it allowed me to be creative while still solving technical problems. I implemented the campus-themed design, improved layouts, worked on navigation flow, and added features such as back buttons and smoother page transitions to improve the user experience. I also worked on improving the listings page by restructuring it into a cleaner two-panel layout so that information was easier to navigate and visually more organized.

Although I did not originally build the sign-in system, I had to spend time understanding how the authentication flow worked so that when I implemented the new homepage and frontend changes, I would not accidentally break existing functionality. This taught me the importance of understanding existing systems before extending them.

One of the biggest challenges I faced throughout the project was figuring out the best way to structure information on a page. I often struggled with deciding where important content should go and how users would naturally interact with the interface. Over time, I learned that frontend development is not only about making things look good, but also about making them intuitive and easy to use.

Outside of frontend work, I also finalized and updated our sequence diagrams so they reflected all the completed functionalities of the system. I additionally maintained our Trello board by ensuring that user stories were clearly written and divided into manageable tasks so the team could stay organized.

One of my proudest moments during the project was working on the homepage animations and visual effects. Specifically, when I imported the student cut-outs in front of the Great Hall and added slight movement to them using React. Seeing my teammates react positively to it made me realize how impactful small UI details can be and helped me appreciate how creative frontend development can be.

Another major learning experience came from working with GitHub workflows and lint testing. At first, it was frustrating when lint tests kept failing during pushes and pull requests, but over time I learned how important testing and clean code practices are. I realized that running "npm test" before committing saves a lot of time and prevents unnecessary debugging later.

If I could restart this project, I would encourage better planning before implementing features, especially features that heavily depend on the database structure. We learned that changing database-related functionality later in development can become very difficult. I would also encourage closer collaboration between frontend and backend developers from

the beginning. Towards the later sprints, pairing frontend and backend members together worked very well because it improved communication, accountability, and progress.

Overall, this project taught me that software development starts with a blank page and endless possibilities. There are so many tools and frameworks available, and each one can help bring ideas to life in different ways. Despite the challenges and frustrations at times, I had an incredible amount of fun throughout this process and gained much more confidence in my technical, creative, and teamwork abilities.